

गण्डकी २०८३

प्रदेश लोक सेवा आयोग, गण्डकी प्रदेश, पोखरा

प्रदेश निजामती तथा स्थानीय सरकारी सेवा अन्तर्गत प्राविधिक तर्फ इन्जिनियरिङ्ग सेवा, सिभिल समूह, जनरल उपसमूह, अधिकृतस्तर सातौं तह, इन्जिनियर वा सो सरह पदको खुला प्रतियोगितात्मक लिखित परीक्षा
विषय: सेवा सम्बन्धी | पत्र: द्वितीय | पूर्णाङ्क: १०० | समय: ३ घण्टा |
मिति: २०८३।०२।३१

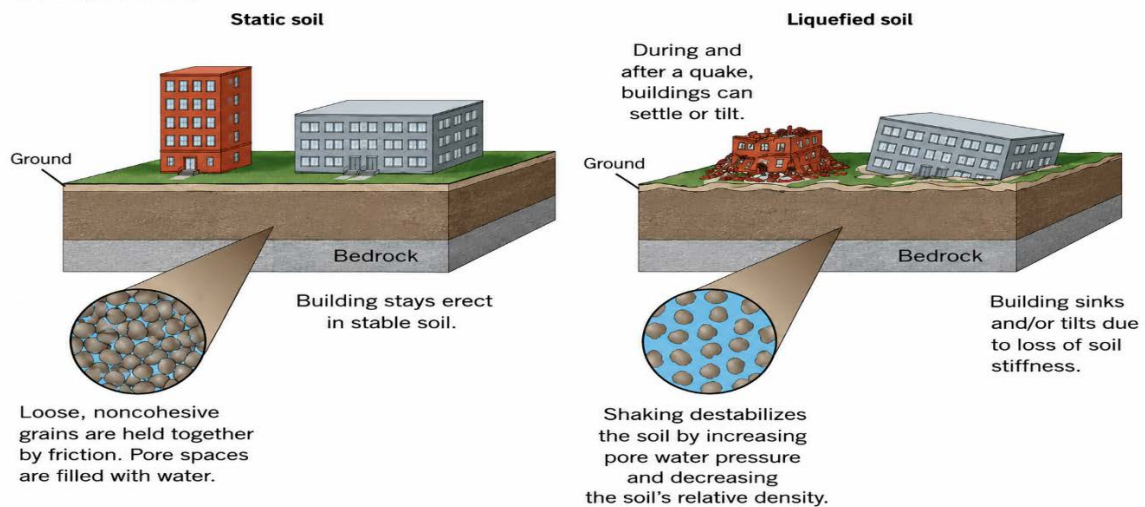
Section-A (25 Marks)

- 1. What is liquefaction of soil? Explain the condition under which liquefaction is most likely to occur. [2+3=5]**

Answer:

Liquefaction of soil is the condition in which the cohesionless soil such as sand and silt, saturated or partially saturated, loses its strength and stiffness due to sudden stress. In this condition, the solid soil behaves like a liquid in response to sudden dynamic forces such as earthquakes and heavy vibrations.

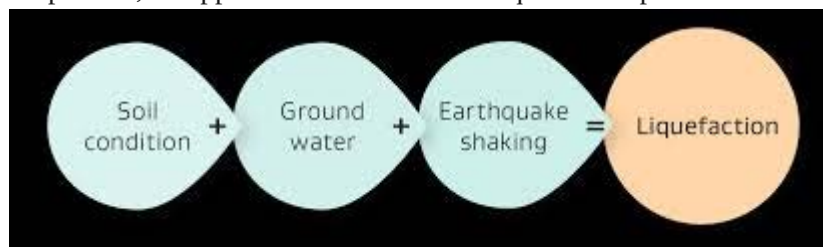
2.4 Liquefaction



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The conditions under which liquefaction is most likely to occur are explained below:

- Cohesionless soil with loose soil structure:**
when the loosely packed soil particles are subjected to sudden dynamic forces then they become more separated supporting to the possibility of liquefaction condition.
- Saturated soil condition:**
The voids between the soil particles must be completely or sometimes partially filled with water to give rise to liquefaction condition.
- Application of Sudden Dynamic forces:**
Earthquakes, heavy vibration due to natural or artificial reasons, when shakes the loosely packed saturated soil particle, it supports the occurrence of liquefaction potential of soil.



- 2. Explain the difference between Working Stress Method and Limit State Method, and discuss why the limit state approach is widely adopted in modern structural design. [3+2=5]**

Answer:

Working Stress Method and Limit State Method are the two methods used for the design of RCC and steel structures. The difference between Limit State Method and Working Stress Method is stated and explained in the points below:

Working Stress Method	Limit State Method
1. Working Stress Method is a traditional and conventional approach of structure design.	1. Limit State Method is a modern approach of structure design.
2. It is a deterministic method which assumes that the stresses due to working load don't exceed the safe permissible stresses.	2. It is a probabilistic approach that considers "Limit States" beyond which the structure ceases to perform.
3. The method is stress-based.	3. The method is strain-based.
4. It utilizes Factor of Safety and produces heavier section.	4. It utilizes Partial Factor of Safety and produces lighter/economical section.
5. Design load is equal to working load in working stress method.	5. Working load is converted into design load by multiplying it with partial FOS.

Limit State Approach is widely adopted in modern structural design because:

- It is more practical, economical and safer approach of design in comparison to traditional Working Stress Method.
- It utilizes two conditions for ensuring the safety of structure till its intended use i.e. Limit State of Safety and Limit State of Serviceability.
- It accounts for uncertainties in loading and material strength separately.

3. Explain the types of bridges and their selection criteria based on span length and site conditions, and discuss in detail the various loads and forces acting on bridges such as live load, impact load, wind load, longitudinal and lateral forces. [5+5=10]

Answer:

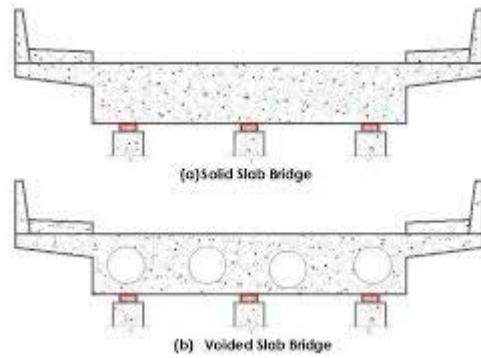
Bridges are the structures providing passage over an obstacle without closing the way beneath it. The obstacle may be river, road, railway, water body or a valley. There are different types of bridges. They are briefly discussed below:

1. According to the materials used for construction of superstructures:

- Steel bridge:
 - made up of steel.
 - Economical and most widely used for both short as well as long spans.
- Timber bridge:
 - made up of wood, timber.
 - Might be used for construction of temporary bridges or at rural areas.
- RCC bridge:
 - made up of Reinforced cement concrete.
 - Strong, durable and most widely used for construction of permanent bridges
- PCC bridge:
 - made up of plain cement concrete.
 - Constructed in the area where traffic load is minimum.
- Composite bridges:
 - made up of different materials such as RCC, timber, steel etc.

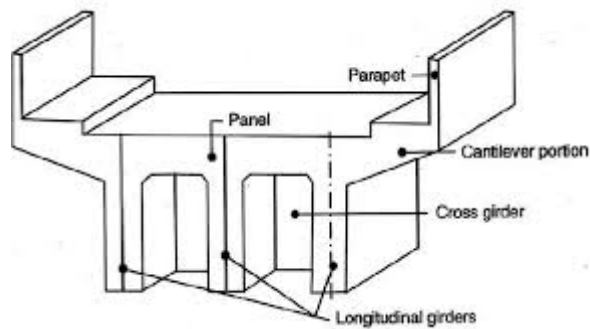
2. According to the types of superstructures:

- Solid slab bridge:
 - Increases clearance above the afflux.
 - Gives clean attractive appearance to the bridge.
 - Economical for the short span length of bridge less than 15m.



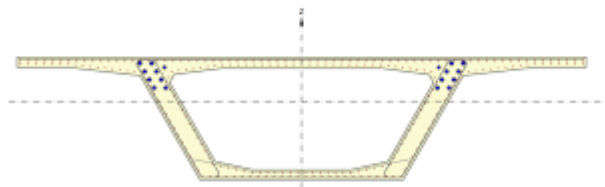
b. T beam bridge:

- Gives rough appearance to the bridge.
- Easy and economical to construct because of monolithic construction of beam and slab.
- Economical for the short span length of bridge less than 30m.



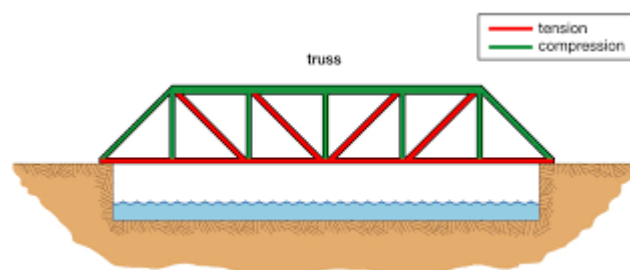
c. Box Girder bridge:

- Gives good appearance to the bridge.
- Reduces the cost of construction for large span.
- Economical for the large span length of bridge greater than 30m.



d. Truss bridge:

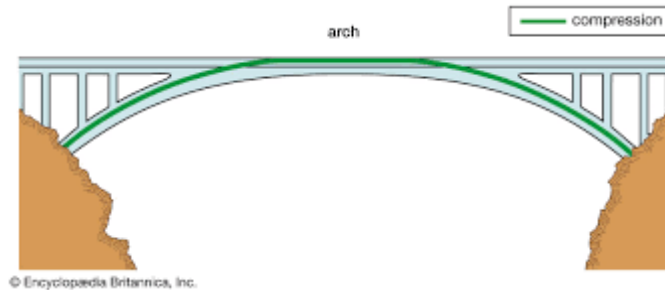
- Easy to construct.
- Used for span length of 50 m to 110 m.
- Requires frequent maintenance and gives rough appearance to bridge.



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e. Arch bridge:

- Gives good appearance to the bridge.
- Economical for the large span length of bridge upto 200m.
- Requires good subsoil condition for foundation.



3. According to span length of bridge:

- Culvert:
 - Used for short span up to the span length of 6m.
- Minor bridge:
 - Used for short span up to the span length of 6m to 30m.
- Major bridge:
 - Used for the span length of greater than 30m.
- Minor bridge:
 - Used for the span length of greater than 100m.

Various loads and forces acting on the bridges are discussed below:

a. Live load:

- Moving load along the length of the road is known as live load.
- According to IRC, live load can be:
 - IRC Class A loading:**
 - Treated as standard loading which is adopted for the design and construction of permanent bridges.
 - IRC Class B loading:**
 - Adopted for the design and construction of temporary bridges and bridges in specifies areas.
 - IRC Class AA loading:**
 - Treated as abnormal loading used for construction of bridges where heavy traffic load is to be subjected.
 - IRC Class 70R loading:**
 - Treated as abnormal loading, which has three patterns of loading i.e. wheeled load, tracked load and train load.

b. Impact load:

- Dynamic effect of moving live load along the length of bridge.
- Impact load is determined by using the formula:

$$\text{Impact load} = \text{Impact Factor} \times \text{Static value of live load}$$

c. Wind load:

- Stress subjected due to wind on the exposed part of bridge.
- $\text{Wind load} = \text{Wind load on the structure} + \text{Wind load on live load}.$
- $F_t = P_z \times A \times G \times C_D$
 Where F_t = Wind load in transverse directions
 P_z = Design wind pressure
 A = Area of the exposed part of structure
 G = Gust Factor

C_D = Coefficient of Drag

d. Longitudinal forces:

- Horizontal forces acting parallel to the length of road.
- Longitudinal forces on the bridge occurs:
 - i. Due to tracking effect through acceleration of driving wheels.
 - ii. Due to braking effect through application of brakes.
 - iii. Frictional resistance offered to the movement of free bearing.

e. Lateral forces:

- In case of highway bridge, lateral load due to parapet and kerb is normally considered.
 - i. Parapet:
Shall be designed to resist lateral horizontal forces and vertical forces each of 150 kg/m applied simultaneously on the top.
 - ii. Kerbs:
Shall be designed for lateral loadings of 750 kg/m applied horizontally on the top of kerbs.

4. Describe the factors affecting permeability of a soil. How do you determine the coefficient of permeability in laboratory and field. [5]

Answer:

Permeability of a soil is the property of soil that allows the water to pass through the voids present in it.

The factors affecting permeability of soil are described below:

a. Void ratio:

- Larger the void ratio, more porous the soil mass which increases the permeability of soil.
- $k \propto e^3 / (1+e)$
-

b. Size of soil particles:

- Coarser the soil particles, larger the size of voids and higher permeability.
- According to Hazen's equation:

$$K = CD_{10}^2 \quad \text{where } C = \text{constant}$$

D_{10} = effective size of particle

c. Shape of soil particles:

- Rounded soil particles have lower specific area and higher permeability while angular soil particles have high specific area and lesser permeability

d. Viscosity of water:

- Viscous liquid reduces the permeability of soil.
- $k \propto 1/\mu$

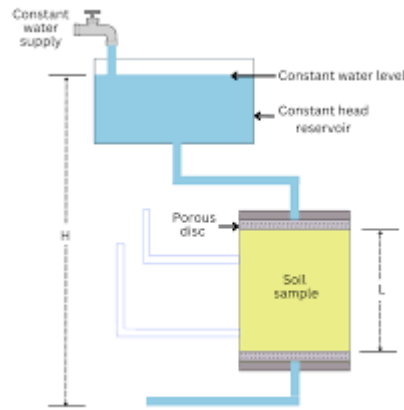
e. Temperature:

- As the temperature increases, viscosity of liquid decreases which increases the permeability of soil.
- $k \propto 1/\eta$

Determination of coefficient of permeability in laboratory can be done by two methods:

i. Constant head test:

- Used for coarse grain soil.
- Test arrangement:



From Darcy's law:

$$q = k i A$$

$$= k \times \frac{h}{L} \times A$$

$$k = \frac{q \times L}{A \times h}$$

ii. Falling head test:

- Used for clay.
- Test arrangement:

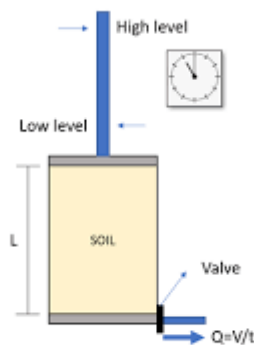


Figure 2: Schematic representation for Falling Head Permeability Method

From Darcy's law:

$$q = k i A$$

$$= k \times \frac{h}{L} \times A$$

$$\frac{dv}{dt} = k \times \frac{h}{L} \times A$$

$$a \times \frac{dh}{dt} = k \times \frac{h}{L} \times A$$

Integrating and solving, we get:

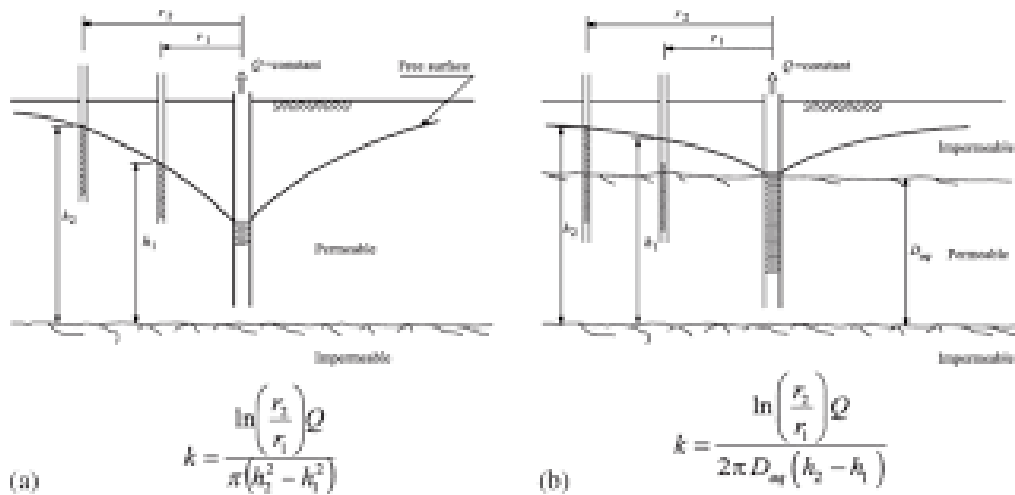
$$k = \frac{2.303 a L}{A t} \log_{10} \frac{h_1}{h_2}$$

Where:

k = coefficient of permeability (hydraulic conductivity)
 a = the inside area of the standpipe
 L = Length of the sample
 A = the inside area of specimen
 t = elapsed time of test
 h_1 = the elevation of water in the standpipe at time $t=0$
 h_2 = the elevation of water in the standpipe at time equal to t .

Determination of coefficient of permeability through field test can be done by two methods:

- a. Unconfined aquifer:
- b. Confined aquifer:



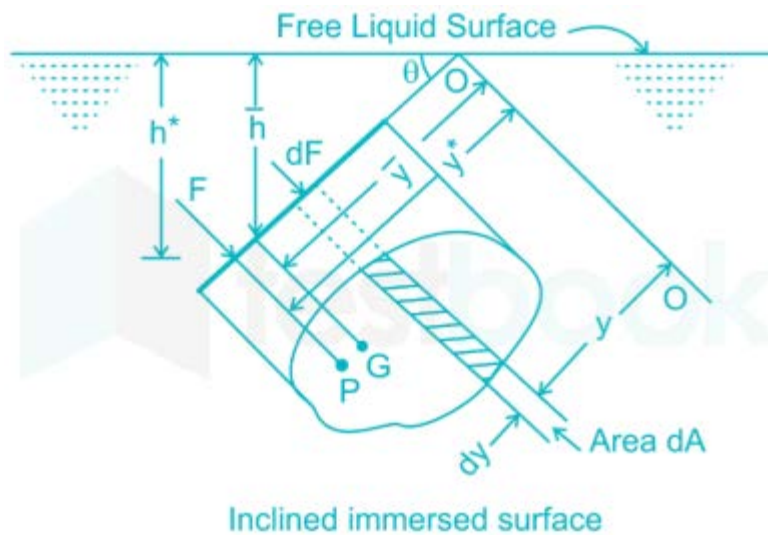
Section-B (25 Marks)

5. Define total pressure and center of pressure in the static liquid. Explain that center of pressure always lies below the center of gravity of the any immersed surface. Take formula without any derivation. [2+3 =5]

Answer:

Total pressure: It is the total resultant force exerted by the fluid on the surface it is in contact with. Also known as hydrostatic pressure.

Center of pressure: It is the specified point through which the resultant pressure acts on that surface.



$$h^* = \frac{(IG \sin^2 \theta)}{A \bar{h}} + \bar{h}$$

For vertical plane surface: $\theta = 90^\circ$

$$h^* = \frac{(IG)}{A \bar{h}} + \bar{h}$$

6. Explain the silt theory and discuss how it influences the design of lined canals in Nepal. [5+5=10]

Answer:

There are two silt theories based on the experiments conducted by respective engineers on the alluvial channel. These theories are used for design of alluvial canals. They are briefly described below.

a. Kennedy's silt theory:

According to Kennedy's silt theory, non-scouring, non-silting velocity is related to flow depth as

$$V_c = 0.55 y^{0.64}$$

He also defined Critical Velocity Ratio (CVR).

$$CVR = \frac{V}{V_c}$$

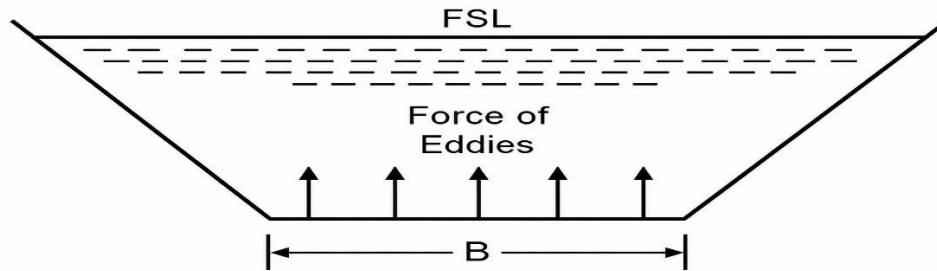
$$V = V_c \times CVR$$

$$V = 0.55 y^{0.64} \times CVR$$

Kennedy found out that $CVR > 1$ (1.1 to 1.2) for sediment coarser than Upper Bari Doab and $CVR < 1$ (0.8 to 0.9) for sediment finer than the canals where he carried out his experiments.

Limitations of Kennedy's theory:

- i. Kennedy only considered average regime.
- ii. No considerations were made for silt grade, silt concentration and bed load.



FSL = Full supply level; B = Width of channel at bed level

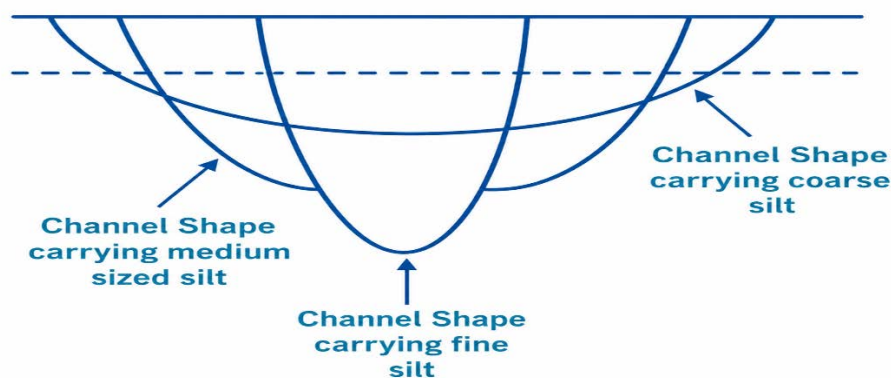
b. Lacey's silt theory:

Lacey's silt theory considered the dimensions of bed width, depth and shape attain a state of equilibrium called regime state.

He also differentiated between the initial and final regime conditions of the channel.

The initial condition of channel is attained shortly after it is put into operation. The canal begins to adjust its bed slope either by silting or scouring. It seems like the canal has reached the stability condition yet is not the final state of stability. This is known as the initial regime condition.

After the continuous action of flowing water upon the canal, it attains the final or true regime condition.



According to Lacey's silt theory,

$$V = (Qf^2/140)^{(1/6)}$$

$$V = 0.48 \sqrt[3]{Rf}$$

$$f = 1.76 \sqrt{d}$$

Silt theories influence the design of lined canals in Nepal fundamentally by:

- i. Determination of flow velocity:

Kennedy's Silt theory has established the relation between average velocity of flow and flow depth from where the velocity of flow can be determined as:

$$V = 0.55 y^{0.64} \times CVR$$

Average flow velocity can also be determined from Lacey's silt formula as:

$$V = (Qf^2/140)^{(1/6)}$$

$$V = 0.48\sqrt{(Rf)}$$

ii. Design of bed width, side slopes and linings of canal:

Design of alluvial canal can be carried out by the help of these silt theories by designing bed width and side slope. For the efficient transport of sediments, canals in Nepal are designed in trapezoidal cross-sections which optimizes hydraulic efficiency of the channel.

iii. Prevention of silting and scouring:

Use of silt theories help the engineers to fix the suitable velocity of flow in the canal to prevent silting. The velocity of flow of water in the canal shouldn't be less than critical velocity which might rise the possibility of silting.

iv. Prevention of scouring:

Silt theories also help the engineers to maintain the average velocity of flow to prevent scouring in the canal. The velocity of flow of water should also not be greater than critical velocity which might rise the possibility of scouring.

v. Management of sediment in canals in different regions of Nepal:

In steep, mountainous region of Nepal, silt theories can be focuses on prevention of scouring. While in the plain terrains of terai region, the focus can be shifted to maintain the velocity such that it prevents silting in the canal.

7. What are the visible impacts of climate change on water resources in Nepal? Mention the measures to mitigate the effects of climate change in drinking water schemes in hills. [5+5=10]

Answer:

Climate change is the global problem however is more impactful in the mountainous country like Nepal. According to German Watch, Nepal ranks 10th among the countries with high risk of climate change. International Panel on Climate Change (IPCC) considers Nepal as one of the most vulnerable countries to climate change with the faster rate of increasing temperature in comparison to the global average temperature. The visible impacts of climate change on water resources in Nepal are described as follows:

i. Glacial Lake Outburst Floods:

Climate change has resulted in melting of the ice caps and glaciers at the increased rate. This results in the formation of unstable glaciers and increases the risk of GLOFs which not only damages the structures at downstream but also creates imbalance in the downstream ecosystem.

ii. Seasonal imbalance:

Climate Change has also contributed to seasonal imbalance in Nepal. Untimely and intensified rainfall causes increased surface runoff resulting to severe flooding. Also, the prolonged dry winter season results water scarcity problems in many areas.

iii. Drying of water resources:

Insufficient and untimely rainfall, prolonged winter season, increased earth temperature has resulted in the drying of traditional natural water resources such as springs and dharas in Nepal.

iv. Increased natural calamities such as floods and landslides:

Increased surface runoff due to untimely and intense rainfall has increased the frequency of the occurrence of natural calamities such as flood and landslides.

v. Water insufficiency:

Climate change has also resulted in water scarcity in many areas for agricultural as well as drinking purposes.

Measures to mitigate the visible impacts of climate change in drinking water schemes in hills are elaborated in the points below:

- i. Construction of catchment ponds:
Catchment ponds can be constructed at different suitable areas to trap the excess surface runoff and utilize the trapped water for groundwater recharge.
- ii. Conservation of forests:
Forests and vegetative covers reduce the surface runoff and facilitates infiltration and percolation of water to the groundwater table. Forests also reduce the velocity of surface runoff and prevent the occurrence of natural calamities such as floods and landslide.
- iii. Water storage infrastructures:
Installation of water storage infrastructures such as multiple elevated water reservoirs to store the excess water during the monsoon season and utilize the water during the dry season, reducing the scarcity of water in the area where availability of water is insufficient.
- iv. Sediment and filtration systems:
Installation of sediment filtration systems at the intake to prevent the entry of debris and sediments in the drinking water to be distributed to consumers.
- v. Proper Utilization of water through Managed Aquifer Recharge (MAR)
Managed Aquifer Recharge is the water management technique in which the stormwater, surface water as well as treated wastewater is directed into the ground for recharging depleted aquifers. This technique enhances natural groundwater recharge.

Section-C (25 Marks)

8. Briefly describe the process of designing drainage structures for Rural Road in hilly region. [5]

Answer:

The process of designing drainage structures for Rural Road in hilly region are stated below:

i. Hydrological analysis:

In hydrological analysis, we determine the maximum runoff that the structure need to intercept.

- a. Collection of rainfall data: Amount, duration, frequency and intensity of rainfall in different time period in a particular area where the drainage structure is to be constructed, must be collected.
- b. Estimation of runoff: Volume of runoff is determined either by using:
 - Rational method (for limited catchment area)

$$Q = CIA/360$$

where:

Q = Peak runoff (m^3/s)

C = Runoff coefficient

I = Rainfall intensity (mm/hr)

A = Catchment area (ha)

- Empirical formula (for large catchment area)

Dicken's Formula:

$$Q = CA^{3/4}$$

where:

Q = Peak runoff (m^3/s)
 A = Catchment area (km^2)
 C = Dicken's coefficient (depends on region)

Ryve's Formula:

$$Q = CA^{2/3}$$

where:

Q = Peak runoff (m^3/s)
 A = Catchment area (km^2)
 C = Ryve's coefficient

ii. Hydraulic analysis:

Hydraulic design of the road side drains can be done by using Manning's equation.

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

where:

V = mean velocity of flow (m/s)
 n = Manning's roughness coefficient
 R = hydraulic radius (m)
 S = bed slope

Discharge:

$$Q = AV = \frac{A}{n} R^{2/3} S^{1/2}$$

where:

Q = discharge (m^3/s)
 A = cross-sectional area of flow (m^2)

iii. Design of cross drainage structures:

Cross drainage structures such as culverts, bridges, causeways are designed and provided at suitable intervals.

iv. Design of subsurface drainage structures:

- a. Depth of trench: It is provided based on the depth at which subsurface flow has to be intercepted.
- b. Selection of backfill materials: Suitable backfill materials provided to improve the interception in the drain.
- c. Holes in perforated drain pipes:
 - Maximum size of circular holes = d_{85}
 - Maximum width of slotted holes = $0.83 d_{85}$
- d. Diameter of pipe: Depends on the amount of water required to be drained off.

9. Explain the classification of highway maintenance activities. Discuss pavement distress types and methods of repair and strengthening of existing pavements. Explain the importance of maintenance planning and inspection in road management. [4+4+2=10]

Answer:

Highway Maintenance activities can be classified into three categories:

i. Routine maintenance activities:

- Those activities that are carried out as frequently as required during each year on all elements of the highway to improve its serviceability in all weather conditions are routine maintenance activities.
- Includes:
 - a. Grass and bush cutting, grading, reshaping of the unpaved surfaces.
 - b. Cleaning of carriageway, drains, slopes, culverts.
 - c. Replacement of structures that has been damaged.



ii. Recurrent maintenance activities:

- Those maintenance activities of localized nature carried out more or less at regular intervals of six months to 2 years are recurrent maintenance activities.
- Includes:
 - a. Sealing cracks, local surface treatment
 - b. Repair of depression and ruts
 - c. local constructions



iii. Periodic maintenance activities:

- Those activities done at the interval of several years to improve the overall functionality of road are periodic maintenance activities.
- Includes:
 - a. Renewal of wearing course of the pavement surface.
 - b. Restoration of road markings
 - c. Repainting of metal bridges etc.



iv. Special maintenance activities:

- Those activities of specialized and emergency nature done to solve special problems which may cause pavement failure if not identified on time.
- Includes:
 - a. construction of temporary diversion to allow the traffic to pass through obstruction.
 - b. Removal of debris and other obstacles.
 - c. Strengthening of pavement structures.



Types of pavement distress can be broadly categorized into two types:

i. Flexible pavement distress:

- a. Alligator cracking:
common type of flexible pavement failure that occurs due to the relative movement of pavement layer materials.
- b. Formation of waves and corrugations:
occurs due to poor subgrade conditions, poor gradation of mix, unstable underlying layers.
- c. Shear failure and cracking:
occurs due to inherent weakness of pavement mix, excessive loading.



ii. Rigid pavement distress:

- a. Scaling of cement concrete:
occurs due to deficiency in the mix or presence of chemical impurities.
- b. Warping cracks:
occurs due to improper design of joints.
- c. Mud pumping:
Soil slurry ejects out through the joints and cracks during the downward movement of slab.



Method of repair and strengthening of existing pavements:

- i. Maintenance of waves and corrugations by proper compaction of subgrade soil.

- ii. Maintenance of joints by cleaning and removal of sealer materials as well as replacement of filler materials.
- iii. Patch repairs: done on the affected localized areas and potholes using a cold mix.
- iv. Base repair: if the failure in the base course layer has been detected then it might be necessary to replace the base course material with new material.

The importance of maintenance planning and inspection in road management are as follows:

- i. Budget optimization: maintenance planning helps in setting priorities and allocate the budget to that pavement requiring prior repair.
- ii. Reduction of hazards: routine inspection helps the authorities to identify and mitigate potential hazards such as potholes.
- iii. Increases lifecycle of the road infrastructures: periodic maintenance and inspection helps to maintain the overall functionality and serviceability of the road which increases the lifespan of the infrastructures.
- iv. Improved travel efficiency: good road condition improves the travel experience to the road users, which enhances the public safety hand in hand.

10. Describe in detail the different criteria used in highway design in Nepal. What are the challenges for selection of road alignment and construction in different geographical regions? [5+5=10]

Answer:

The different criteria used in highway design in Nepal are described in detail below:

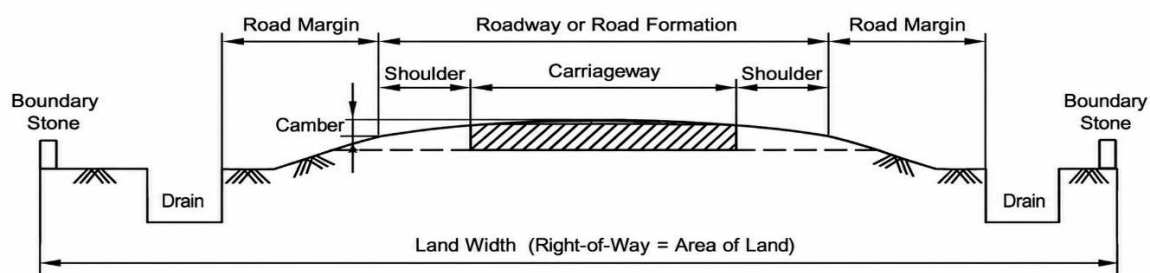


FIG. 1 CROSS-SECTION DETAILS OF A ROAD

i. Design speed:

- Most important factor affecting the geometric design of highway.
- Design speed is the highest speed at which vehicle can travel safely in the highway under favorable climatic conditions.
- Affects the sight distance, horizontal distance, length of horizontal and vertical curves.

Terrain Classification and Design Speed Tables

Terrain classification	Cross slope (%)
Plain	0–10
Rolling	10–25
Mountainous	25–60
Steep	60

The recommended design speed is given in Table (as per IRC – ruling and minimum).

Road Class	Plain	Rolling	Hilly	Steep
Class I	120	100	80	60
Class II	100	80	60	40
Class III	80	60	40	30
Class IV	60	40	30	20

ii. Design vehicle:

- Dimensions, weight of axle, operating characteristics of vehicles influence the aspects of geometric design such as pavement width, radii of curvature, clearance distances etc.
- According to NRS 2070,
Maximum width = 2.5 m'
Maximum height = 4.75m
Maximum length = 18 m
Maximum single axle load = 100 KN

iii. Topography:

- Geometric design standards are different for different terrain to keep the cost of construction and time of construction under control for economical design and operation of highway.
- Terrain can be plain, rolling, mountainous, steep depending upon the average cross slope of degree of slope, which affects the design speed ultimately influencing the geometric design on highway.

iv. Traffic factors:

- Traffic factors such as human characteristics and vehicular characteristics affect the design of geometric elements of the highway.
- The human characteristics that influence geometric design are physical, mental and psychological characteristics.

v. Design hourly volume and capacity:

- As it is uneconomical to design the road for peak traffic flow, 30th highest hourly volume is used for design purpose.
- Traffic capacity must be estimated to determine the number of lanes required which helps to allocate the width of the road.

The challenges for selection of road alignment and construction in different geographical regions are mentioned in the points below:

i. In mountainous and hilly regions:

- Slope instability:**
Excavation and construction of road in mountainous regions might induce the challenges of rockfall and landslides due to unstable slope.
- Drainage problems:**
Due to steep topography in mountain and hilly regions, high speed runoff must be properly intercepted by providing suitable drainage works which is often costly and difficult to provide in such regions.
- Harsh climatic conditions:**
Extreme climatic conditions such as lower atmospheric pressure, freezing temperatures in mountainous regions make it difficult for conducting road pavement works, construction and maintenance works.
- Difficult geometry and road alignment:**
Inclusion of hairpin bends in the suitable location, balancing cut and fill volumes are some of the major challenges in selection, design and construction of road in mountain and hilly regions.

ii. In terai region:

- Flooding problems:**
One of the major challenges for selection of road alignment and construction in terai region is flooding. Hence, to mitigate this problem suitable drainage provisions must be made.

- b. Poor soil conditions:
Poor subgrade soil conditions in water logged areas, marshy lands etc. lacks the bearing capacity to withstand the subjected traffic volume due to which pavement failures, settlements can be seen.
- c. River morphology:
Heavy silting in the river, erosion of side banks, shifting its course are very common in the slow flowing rivers of terai region. So, selection of road alignment and construction can be quite challenging in the highly dynamic floodplains.
- d. High Groundwater table:
High level of GWT affects the quality and performance of subgrade soil which ultimately affects the performance of the road. Hence, selection of road alignment and construction must be done properly by installation of suitable drainage facilities as required.

Section-D (25 Marks)

11. Discuss the concept of 'Greywater Management' in sustainable sanitation. Explain its role in reducing fresh water demand in water scarce rural areas and outline low-cost reuse options suitable for Nepali households. [3+3+4=10]

Answer:

Greywater/sullage is the domestic wastewater obtained from wash basins, kitchen sinks, bathroom which has no organic matter contained in it. This greywater can be collected, recycled and reuse to reduce the fresh water demand using simple treatment operations which is known as greywater management.

The collected greywater is used to irrigate vegetables in home and kitchen gardens especially during the dry season when water sources dry up. It is a low cost, effective and useful intervention that can be easily adopted in drought prone and water-scarce areas to cultivate vegetables for family nutrition, income and livelihood of small-scale farmers. Hence, Greywater is mostly used for irrigation, toilet flushing purposes and other minor uses of water and reduces the reliability on potable water sources for non-potable use of water.

In the context of Nepal, greywater management is being practiced in different areas of rural hilly region where water is scarce, under the initiative of community-based organizations such as Sahas Nepal. This organization is working to promote greywater management technology in its working areas. Another organization known as smartpaani is also working in the field of greywater and wastewater management in Nepal to save water and reduce the wastages of potable water for non-potable use.

Advantages of greywater management:

- i. Conservation of water:
Utilization of greywater for irrigation and flushing helps to conserve fresh water and reduce the demand of potable municipal water by 30-45%.
- ii. Reduction of load on treatment plants:
Diverting and reusing greywater minimizes the volume of wastewater that reaches the treatment plant which helps to reduce the organic load on treatment units.
- iii. Over encroachment of water resources:
Fulfilling the minor water demands by utilizing greywater reduces the over-use of natural resources and prevents its over encroachment.

Role of greywater management in reducing fresh water demand in water scarce rural areas:

- i. Substitution of potable water for non-potable uses:

Greywater management helps in replacing the potable water for its use in non-potable areas such as irrigation, livestock, toilet flushing. Utilization of greywater by reusing and recycling it, minimizes the wastage of fresh water and scarcity of water to great extent.

- ii. Groundwater recharge:
Greywater can be collected, treated and safely filtered back to the soil. This helps to prevent stagnation of water and also replenish the local shallow aquifers which is beneficial to the water scarce rural areas to overcome the scarcity of water.
- iii. Conservation of ground and surface water resources:
Greywater management reduces the reliability of consumers to the potable water resources and its over encroachment. Hence, it is one of the effective techniques of conservation of water resources.

Some of the low-cost reuse options suitable to Nepali households:

- i. Rainwater harvesting:
 - It is the process in which rainfall runoffs from the roofs, yards, open spaces are collected, filtered, treated and utilized for future use.
 - It is used for household chores, irrigation, livestock, groundwater recharge and other non-potable utilization of water.
 - Rainwater harvesting is simple, effective, economical and sustainable process of recycling and reusing water that helps to reduce reliability on potable sources for non-potable use of water and prevents water scarcity in the area.
- ii. Greywater management:
 - It is the process of collecting, recycling and reusing the greywater obtained from bathrooms, kitchen sinks, wash basins to meet the demand of water for non-potable use in households.
 - It helps in conservation of water resources and prevent over encroachment and scarcity of water.
- iii. Reuse of plastic containers:
 - Plastic containers instead of discarding can be reused for storage of grains or other household items, vertical gardening systems, home made drip irrigation for kitchen gardening, seedling nurseries and so on.
- iv. Biogas plants:
 - Organic wastages such as human excreta, faeces of animals can be utilized by converting it into cooking gas by the help of biogas plants.
 - One of the most popular methods of sewage disposal and practiced in terai region of Nepal.
- v. Organic Fertilizers:
 - Organic wastages can also be converted into soil conditioners and fertilizers useful for the growth of plants, by utilizing nitrogenous and phosphorus contents in them.

12. Draw a flowchart of each component of typical Gravity water supply system in rural area of Nepal. Explain the function of each component in brief: [5+5=10]

Answer:

In a gravity water supply system in rural hilly area of Nepal, source is usually fast flowing rivers and springs which is relatively free from harmful suspended materials. Hence, treatment plant is not incorporated in water supply project in rural hilly areas. However, in a gravity water supply system in rural terai regio, treatment plant must be incorporated to remove the suspended and other harmful materials from the water drawn from the source.

A flowchart of typical gravity Water Supply System (RWSS) in rural area of Nepal is shown below:

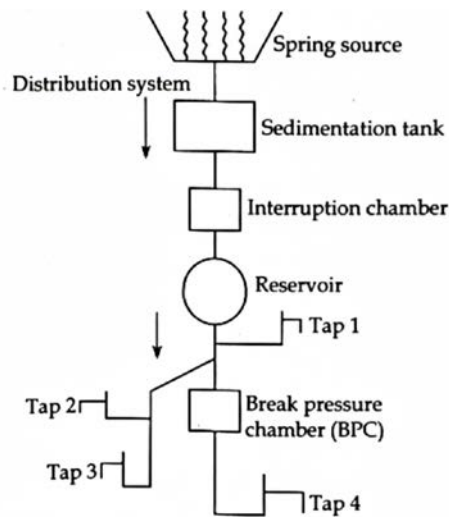


Figure: Typical rural water supply system

Components of water supply systems:

- i. Source of water:
 - Source of water should be reliable and contain minimum impurities.
 - The amount of water available in the source must be adequate to tap the required amount to supply it to consumers.
 - For rural hilly areas, source of water is springs which is considered to be relatively pure hence no treatment plant is provided in the water supply project of such areas.
- ii. Intake:
 - It is a device or structure constructed at or near the water source to draw the required amount of water to supply line.
 - Must be easily accessible and must have minimum possibility of contamination.
 - Primary function of intake is to draw the required amount of water from source.
- iii. Distribution chamber:
 - The main function of distribution chamber is to facilitate the divisions of flow in distribution subsystems.
- iv. Transmission mains:
 - The main function of transmission mains is to convey the water from source to treatment plant.
- v. Interruption chamber:
 - Interruption chamber is provided in transmission mains to break the pressure without float valve to prevent bursting of pipe due to excess pressure.
- vi. Reservoir:
 - The main function of reservoir is to store water from the source during low demand period and utilize it during high demand.
- vii. Valve chamber:
 - Chamber constructed to protect the valve in water supply projects.
- viii. Distribution system:
 - It is the pipe network to deliver water to consumer from the distribution reservoir.
- ix. Break pressure tank:
 - It is a pressure releasing device to break the pressure with float valve to prevent bursting of pipe due to excessive pressure.
 - Provided in distribution mains.